

SHABLA, Bulgaria

WMO: 155610

Lat: 43.533N

Lon: 28.533E

Elev: 28

StdP: 100.99

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.7	-11.9	1.4	-8.5	-9.7	1.7	-6.0	16.4	-3.7	16.1	-4.4	9.3	320	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	29.0	24.0	28.0	23.1	27.1	22.5	25.1	28.0	24.1	27.0	23.2	26.3	5.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.0	27.6	23.1	17.9	26.7	22.1	16.8	25.9	76.8	28.0	72.8	27.1	69.2	26.4	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
14.4	12.1	10.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-11.7	31.7	3.4	2.1	-14.1	33.2	-16.1	34.4	-17.9	35.6	-20.4	37.2		
			-11.9	26.0	3.2	0.9	-14.3	26.7	-16.2	27.3	-18.0	27.8	-20.3	28.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	2.2	3.0	5.9	9.6	14.9	19.6	22.5	22.7	18.6	14.0	8.9	4.0	(6)
(7)		DBStd	8.06	4.70	4.43	3.56	2.78	2.71	2.37	2.10	2.10	2.73	3.68	4.19	4.85	(7)
(8)		HDD10.0	889	244	198	136	39	2	0	0	0	11	69	192		(8)
(9)		HDD18.3	2602	501	429	387	263	111	14	1	29	139	284	444		(9)
(10)		CDD10.0	1691	1	1	7	27	152	288	387	394	259	134	35	6	(10)
(11)		CDD18.3	364	0	0	0	0	2	52	129	137	39	4	0	0	(11)
(12)		CDH23.3	2225	0	0	1	0	4	214	913	1001	88	4	0	0	(12)
(13)		CDH26.7	288	0	0	0	0	0	12	127	147	2	0	0	0	(13)
(14)	Wind	WSAvg	5.4	5.9	6.1	5.8	5.0	4.6	4.6	4.7	4.9	5.4	5.8	5.7	6.2	(14)
(15)	Precipitation	PrecAvg	454	30	22	31	33	35	35	39	33	68	47	38	42	(15)
(16)		PrecMax	705	82	62	94	122	82	98	177	104	295	234	93	146	(16)
(17)		PrecMin	239	3	2	3	2	6	0	2	1	5	0	6	4	(17)
(18)		PrecStd	125	20	15	20	26	21	25	40	31	65	45	23	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.6	18.2	20.2	23.3	28.1	30.3	30.6	26.5	23.6	19.0	14.9	(19)
(20)			MCWB	10.0	9.8	11.8	14.5	18.4	22.9	23.2	24.7	20.2	19.4	15.3	12.5	(20)
(21)		2%	DB	10.9	11.3	14.2	17.1	21.5	26.5	28.9	29.4	25.2	21.8	17.3	13.4	(21)
(22)			MCWB	8.9	8.2	9.6	12.7	17.8	22.0	23.9	24.4	20.5	18.6	15.1	11.5	(22)
(23)		5%	DB	9.5	9.5	11.9	14.9	20.5	25.5	28.0	28.2	24.3	20.7	15.9	12.1	(23)
(24)			MCWB	7.8	7.4	8.8	12.0	17.4	21.5	23.2	23.4	20.1	17.9	13.9	10.5	(24)
(25)		10%	DB	8.3	8.4	10.5	13.6	19.6	24.6	27.2	27.2	23.5	19.5	14.6	10.9	(25)
(26)			MCWB	6.9	6.6	8.1	11.3	16.7	20.8	22.8	22.6	19.7	16.9	12.8	9.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.2	12.1	15.1	19.5	24.1	25.9	26.6	22.5	20.6	16.5	12.9	(27)
(28)			MCDWB	12.9	13.7	17.8	18.4	21.7	26.7	28.6	29.6	25.1	22.7	17.9	14.3	(28)
(29)		2%	WB	9.0	8.6	10.1	13.4	18.6	23.0	25.1	25.4	21.6	19.3	15.1	11.8	(29)
(30)			MCDWB	10.6	10.8	13.2	16.3	20.8	25.5	27.9	28.6	24.2	21.2	16.9	13.2	(30)
(31)		5%	WB	8.0	7.6	9.2	12.3	17.7	22.1	24.2	24.3	21.0	18.2	14.1	10.6	(31)
(32)			MCDWB	9.3	9.3	11.5	14.5	20.0	24.9	27.2	27.3	23.6	20.2	15.7	11.9	(32)
(33)		10%	WB	6.9	6.8	8.4	11.5	17.0	21.2	23.3	23.4	20.2	17.1	13.1	9.4	(33)
(34)			MCDWB	8.2	8.3	10.3	13.4	19.3	24.0	26.5	26.5	22.9	19.2	14.6	10.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	5.7	5.8	5.9	6.6	7.3	7.7	7.9	7.0	6.1	5.9	5.9	(35)
(36)			MCDBR	6.9	7.5	8.4	8.2	7.8	8.5	8.6	8.8	7.9	7.2	6.4	6.2	(36)
(37)			MCWBR	5.3	5.3	5.2	5.5	5.6	6.0	6.0	6.3	5.5	5.4	4.8	5.0	(37)
(38)		5% WB	MCDWR	6.3	6.9	7.2	6.9	7.0	7.5	7.4	7.5	6.5	6.3	5.9	5.5	(38)
(39)			MCWBR	5.2	5.1	4.8	5.0	5.4	6.1	6.1	6.1	5.3	5.2	4.8	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.358	0.387	0.417	0.458	0.430	0.432	0.441	0.451	0.431	0.400	0.377	0.360	(40)	
(41)		taud	2.324	2.243	2.187	2.088	2.258	2.292	2.278	2.242	2.266	2.321	2.322	2.322	(41)	
(42)		Ebn at Noon	749	788	813	808	845	843	830	807	791	767	719	705	(42)	
(43)		Edh at Noon	85	108	129	154	133	129	130	131	118	99	84	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.31	2.07	3.37	4.88	6.20	6.78	6.75	6.05	4.31	2.69	1.57	1.18	(44)	
(45)		RadStd	0.20	0.25	0.35	0.47	0.34	0.41	0.37	0.29	0.36	0.25	0.20	0.19	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-114	+105	+61

Nomenclature: See separate page